

Towards a Systematic Mapping Review of Sustainable HCI

Opening Up Our Protocol to the Feedback of the CHI Community

Jordi Márquez Puig
Playful Living Lab
ERAM Escola Universitària de les Arts
Salt, Girona, Spain
jordi.marquez@eram.cat

Daniel Fernández Galeote
Gamification Group, eResearch Centre
of Gameful Realities
Tampere University
Tampere, Finland
daniel.fernandezgaleote@tuni.fi

Anton Bardera Reig
Graphics and Imaging Laboratory
Universitat de Girona
Girona, Spain
anton.bardera@udg.edu

Juho Hamari
Gamification Group, Research Centre
of Gameful Realities
Tampere University
Tampere, Finland
juho.hamari@tuni.fi

Ferran Altarriba Bertran
Playful Living Lab
ERAM Escola Universitària de les Arts
Salt, Girona, Spain
Gamification Group, Research Centre
of Gameful Realities
Tampere University
Tampere, Finland
ferran.altarriba.bertran@eram.cat

Abstract

This Poster presents the design and current state of an ongoing Systematic Mapping Review (SMR) of Sustainable Human–Computer Interaction (SHCI) research published in ACM SIGCHI venues between 2007 and 2025. Rather than reporting consolidated findings, we open the foundations of our review process to the CHI community’s feedback. We share (1) the protocol we used to construct and screen the corpus, (2) a set of analytical lenses we aim to use to guide our analysis, and (3) an early-stage visualization platform to explore and interrogate the mapping of the analyzed data. By making these elements explicit before analyze the literature, we invite the SHCI community to assess corpus adequacy, question assumptions embedded in the lenses of analysis, and suggest meaningful visualizations of the data that respond to the community’s needs and support collective sense-making of the field, in the face of climate change and the need for sustainability transformations.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models.**

Keywords

Sustainable HCI, systematic mapping review, climate change, sustainability, visualization tools

ACM Reference Format:

Jordi Márquez Puig, Daniel Fernández Galeote, Anton Bardera Reig, Juho Hamari, and Ferran Altarriba Bertran. 2026. Towards a Systematic Mapping Review of Sustainable HCI: Opening Up Our Protocol to the Feedback



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3799275>

of the CHI Community. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3799275>

1 Introduction

Since the articulation of Sustainable Interaction Design by Blevis[3], Sustainable Human–Computer Interaction (SHCI) has grown into a broad and heterogeneous field, spanning diverse technologies, domains, and scales of intervention. Prior reviews and critical analyses have offered valuable perspectives on SHCI[5, 7, 8], but they often operate within specific analytical frameworks and leave the boundaries of the field implicit; for example, by focusing on critical reflections of the SHCI landscape[4], exploring specific technological domains such as virtual and augmented reality[6], or looking into particular target audiences and intervention strategies[1].

Several authors have critically noted that the absence of a clearly articulated and shared vision for sustainability within HCI can limit the field’s capacity to orient research agendas or support collective action[10, 15]. For example, certain dominant framings (such as narrowly persuasive approaches) risk reducing sustainability to a motivational device rather than a site of deeper theoretical and methodological engagement[5]. As a result, it often becomes difficult to assess what SHCI contributes, as a field, to sustainability research, or how different approaches relate to one another. As a response, Knowles et al. draw on Naomi Klein’s argument that climate change can provide a much-needed narrative clarity, capable of unifying otherwise fragmented or seemingly disconnected activities around a shared project of societal transformation[9].

To address the gap of a much needed holistic sense-making of SHCI as a whole and to create a systematic canvassing of the current state of sustainability research in HCI we are working on a Systematic Mapping Review (SMR) of this research area. In this poster paper, we present our ongoing efforts toward producing the SMR. We are inspired by previous work that treats literature mapping as

an exploratory and reflective practice that benefits from discussion among the research community it explores[2]; consequently, rather than sharing a finalized analysis, we open up the foundations of our review early in the process, before the actual analysis has been done, so the SHCI community can contribute to strengthening them. In particular, we share our process for collecting relevant literature, the analytical framework we intend to use, and our early ideas for visualizing the results of the analyzed literature. We intend to discuss these three items at CHI 2026, using our poster as an anchor, hoping that it will strengthen the foundations of our SMR and its usefulness for the SHCI community.

2 Current State of our Systematic Mapping Review

To landscape the current state of SHCI research, we decided to undertake a Systematic Mapping Review (SMR)[12, 13]. Systematic mapping reviews aim to provide a structured and exploratory overview of a research field by classifying and organizing a broad body of literature to identify patterns, gaps, and research trends, rather than synthesizing evidence or evaluating effectiveness. In this poster paper, we report on the current state of our review, which is still at an early stage of completion, with the explicit intention of opening its foundations to community validation and feedback. At the moment of submitting this paper, our work has focused on the following areas:

- **Protocol and corpus construction.** We have defined and operationalized a review protocol to construct a corpus of SHCI literature published within the ACM SIGCHI ecosystem (data source, temporal scope, search strategy, screening criteria), and applied it to build an initial analytical corpus. We describe the protocol in Section 3.1. At CHI, we will seek community input to validate its scope and to identify relevant venues, publication types, or sub-communities that may be missing or underrepresented in our dataset.
- **Analytical lenses.** We have developed a set of lenses to examine SHCI literature across multiple dimensions, including: research contributions, sustainability framings, human activities, technologies explored, target users, and roles attributed to technology. We describe them in Section 3.2. These lenses are provisional rather than finalized – we invite feedback on adequacy, overlaps, omissions, and assumptions, as well as proposals for additional lenses or alternative framings.
- **Visualization and analysis platform.** In parallel, we are developing a custom platform to support the SMR process by enabling exploration of the corpus through combinations of lenses and categories, supporting visualizations and cross-analyses that reveal patterns, gaps, and tensions. We describe it in Section 3.3. The platform is at an early stage (*Project website*: <https://smr.playful.eram.cat>); we therefore invite brainstorming on which visualizations, interactions, and analytical views would be most valuable for examining and debating the SHCI landscape.

All coding and analytical decisions within the SMR will be conducted by the authors without the use of AI-based tools for classification or interpretation, to preserve methodological transparency, interpretive accountability, and consistency across the corpus. At

the same time, we are reflecting on how the broader SHCI community might be meaningfully involved in the process. While this paper outlines initial ideas for community engagement—such as feedback on the protocol, lenses, and visualizations—these mechanisms are intentionally provisional. We expect that the format and extent of community participation will evolve through the discussions that take place around the poster at CHI, and we remain open to collectively shaping how such involvement should be structured.

3 Review Design & Scope

We adopt an SMR approach to characterize SHCI in a structured and transparent manner. Rather than synthesizing evidence around a narrow question, the review seeks to map a broad and heterogeneous corpus of literature to identify patterns, gaps, and tensions over time. The design of our SMR protocol combines replicable procedures (e.g. and explicit corpus construction and screening) with an open and reflexive stance. Here we present our key methodological choices to enable community discussion and critique.

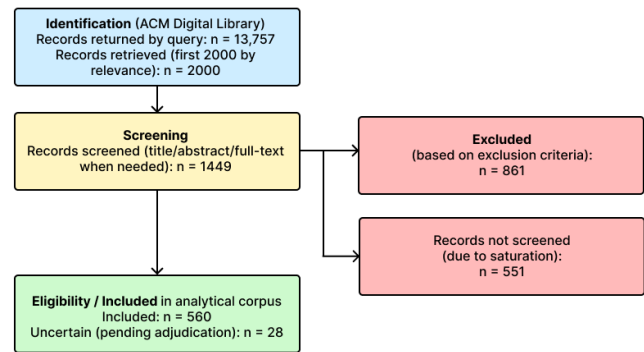


Figure 1: Flow diagram of the corpus identification and screening process for the SHCI SMR. The diagram is adapted from PRISMA 2020 guidelines to reflect the exploratory and mapping-oriented nature of the review.

3.1 Corpus Definition and Construction

We constructed the corpus from the ACM Digital Library, focusing on ACM SIGCHI venues between 2007 and 2025. To maximize recall, we queried sustainability-related terms across all indexed fields.

- AllField: (sustainability OR sustainable OR "climate change")

and applied the following filters:

- Source: ACM SIGCHI venues
- Publication date: 01/01/2007 – 31/12/2025

We established 2007 as the starting point in correspondence with the publication of Eli Blevis' foundational work on Sustainable Interaction Design[3], which marks the explicit emergence of sustainability as a central concern within HCI. We treat the resulting corpus as a provisional delimitation of SHCI within the SIGCHI ecosystem, and open it to revision based on community feedback (at CHI and beyond). We conducted the screening process drawing on selected PRISMA principles[11], adapted rather than strictly followed, to support transparency and traceability in an exploratory

SMR context. We applied exclusion criteria progressively (title and abstract, and, when needed, full text):

- DP – Duplicated papers.
- LC – Non-English publications.
- NF – No full text available.
- NR1 – Not related to environmental sustainability.
- NT – Not eligible publication type (e.g., Doctoral Consortium, Student Design Competition, Workshop Papers, Keynotes, Tutorials, Columns/Forums/Opinion Pieces).

We present these criteria as provisional methodological choices rather than neutral filters, and we keep them open to discussion as part of refining the corpus prior to final synthesis. The initial ACM DL query returned 13,757 records; due to interface constraints of the ACM Digital Library, only the first 2,000 results ordered by relevance are accessible for screening. We therefore applied our inclusion and exclusion criteria to these records. Of these, we screened 1,449 articles, yielding 560 included papers, 861 exclusions, and 28 uncertain cases pending clarification.

We proceeded with the screening in descending relevance order as ranked by the ACM DL. In the final tranche of 50 screened articles, only 8% met the inclusion criteria, and most records could be excluded straightforwardly based on the exclusion codes above. We interpret this declining inclusion yield as an indicator of screening saturation within this source, while keeping open the possibility of extending or revisiting the corpus based on community feedback.

3.2 Analytical Lenses

To analyze the literature, we devised a set of analytical lenses that make explicit how SHCI conceptualizes sustainability, positions technology, and articulates mechanisms of change. Each lens addresses a specific analytical question and combines deductive and inductive coding where appropriate, aiming to surface assumptions that often remain implicit in SHCI reviews. Here we present these lenses as a provisional framework we would like to discuss, modify, and extend through conversation with other researchers at CHI.

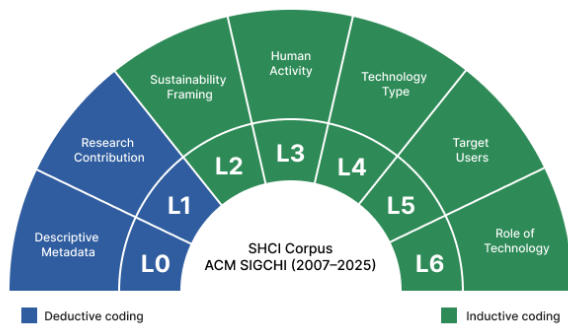


Figure 2: Semi-circular overview of analytical lenses and coding logic used in the SMR. Visual encoding indicates deductive and inductive approaches across lenses.

- **Lens 0 – Descriptive Metadata:** This lens establishes a shared descriptive baseline for the entire corpus. It includes metadata such as publication year, venue (conference or

journal), type of paper (e.g., full paper, notes, poster), and other relevant institutional indicators. The purpose of this lens is not interpretive, but rather to situate SHCI research temporally and institutionally. It enables the observation of patterns in the evolution of the field, concentrations across venues, and shifts in the types of contributions over time. This information serves as a baseline for contextualizing the remainder of the analysis.

- **Lens 1 – Research Contribution:** To analyze the type of contribution each article makes to HCI, this deductive lens is based on the taxonomy proposed by Wobbrock and Kientz[16], which identifies several types of research contributions, including empirical research, artifact creation, methodological development, theoretical contributions, dataset provision, surveys, and critical analyses. This lens enables observation of what SHCI produces as a field: systems and prototypes, empirical knowledge, conceptual frameworks, critique, and reflection, among others. Importantly, a single article may be coded with more than one contribution type, acknowledging the hybrid nature of much HCI research.
- **Lens 2 – Sustainability Framing:** This lens explores how sustainability is understood and framed in each article. Because SHCI research is diverse, we do not begin with a fixed definition or predefined categories. Instead, we use an inductive approach, allowing different sustainability framings to emerge from the articles themselves, based on how authors define, apply, or question the concept. This makes it possible to see whether sustainability is treated mainly as an environmental issue, or whether it also includes social, behavioral, cultural, economic, or systemic dimensions. In this way, the lens does more than identify themes — it also helps reveal how sustainability is conceptually constructed, limited, or implicitly assumed within SHCI research.
- **Lens 3 – Human Activity:** This lens examines sustainability through domains of everyday human activity, drawing on practice-theoretical accounts that conceptualize sustainability as embedded in routines, practices, and sociomaterial arrangements rather than isolated individual behaviors[14]. Within SHCI, this perspective has informed a sustained focus on everyday activities such as energy use, mobility, food practices, consumption, and domestic life[7]. Rather than relying on a predefined taxonomy, this lens adopts an inductive approach where activity categories are developed through ongoing engagement with the corpus. Initial orientations drawn from practice theory and prior SHCI work serve as starting points rather than fixed classifications, allowing categories to be refined, expanded, or reconfigured as the analysis unfolds.
- **Lens 4 – Type of Technology:** This lens analyzes the types of technologies that are proposed or studied in the corpus. We adopt an inductive approach to avoid imposing closed categories (e.g., AI, IoT, mobile apps, etc) and thus allow technological typologies to emerge from the articles themselves. In this way, the lens reflects the diversity of technological approaches present in SHCI, including digital artifacts, hybrid systems, platforms, or material devices, without presupposing which technologies are most relevant a priori.

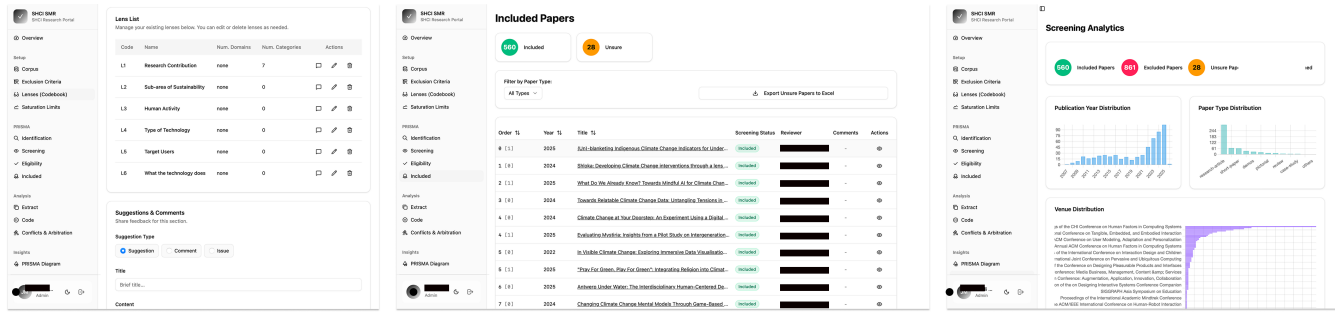


Figure 3: Screenshots of the custom web-based platform developed to support the SHCI Systematic Mapping Review. The interface enables management of analytical lenses, screening and inclusion decisions, and exploratory visual analytics, and is intended as a shared research tool for collaborative analysis and community feedback.

- **Lens 5 – Target Users:** This lens examines who the technology or research is addressed to, such as individuals, households, communities, professionals, organizations, or other actors. We adopt an inductive approach to allow categories of target users to emerge from the corpus itself, based on how responsibility and agency are articulated in the reviewed articles. This lens is particularly relevant for identifying how responsibility for sustainable change is distributed. It enables the detection of whether interventions focus primarily on the individual or whether they incorporate collective, institutional, or infrastructural scales, thereby supporting a critical reading of the implicit assumptions present in the field.
- **Lens 6 – What the technology does:** Finally, this lens analyzes the role attributed to technology within sustainability agendas. We adopt an inductive approach to capture the diversity of roles technologies are assumed to play, rather than imposing predefined functional categories. The aim is to understand what the technology is expected to do, such as informing, persuading, facilitating practices, coordinating actors, raising awareness, supporting decision-making, or provoking critical reflection, among many others. This lens explicitly captures the assumed theory of change embedded in each article, highlighting how technology is conceived as an agent of sustainable transformation. In this sense, it is one of the most characteristic lenses of SHCI and a central element for critical discussion of the field.

3.3 Visualization Platform as a Research Tool

We developed the visualization platform not only as technical support for analysis, but as a research artifact that helps surface methodological assumptions in SHCI mapping. It will enable interactive exploration of lens combinations, supporting cross-analyses that can reveal patterns, gaps, and tensions across the corpus.

Beyond visualization, we designed the platform to support community participation by enabling (1) corpus expansion through proposals of missing papers or venues, and (2) collaborative refinement of lenses through discussion of edge cases and disagreements. In this sense, we conceived this tool as a shared infrastructure for collaborative SMR work in SHCI, not only as a way to access results.

The visualization component of the tool is currently at an early stage of development. Our initial approach is to conceive it as a flexible environment for interactive exploration of the corpus, in which users can combine analytical lenses and categories, select different types of visual representations, and articulate the analytical questions or hypotheses they wish to explore through the mapping. As we move forward with its development, we would like to understand how this kind of visualization tool might be useful for (and, in practice, leveraged by) the community. Thus, we invite other SHCI researchers to propose (1) which combinations of lenses and categories they would find most meaningful to explore, (2) which types of visualizations could best support such explorations, and (3) what kinds of questions or insights they would expect a mapping of SHCI to help address.

4 Conclusion & Call for Feedback

We are working toward producing a Systematic Mapping Review (SMR) of Sustainable HCI literature. In this paper, we presented our ongoing efforts toward that end, opening the early foundations of our review to the scrutiny of the CHI community. We explicitly position community engagement as a core methodological step in our process, rather than a post-hoc validation, as we believe is essential to make our review resonate with the needs of our community. At CHI, we will intentionally design both the poster and the conversations around it to foster critical dialogue, treating the poster session itself as a methodological space where the foundations of our review can be fleshed out, contested, and collectively shaped.

We invite feedback on three interconnected aspects of our review: (1) the scope and composition of the corpus, including potential omissions or underrepresented SHCI sub-communities; (2) the proposed analytical lenses, their adequacy, overlaps, and underlying assumptions; and (3) the design of the visualization platform, particularly regarding which forms of data visualization and cross-lens analysis could best support collective sense-making of the field.

Beyond this specific review, we propose that the analytical and visualization tool could function as a shared infrastructure for collaborative analysis, enabling multiple researchers to contribute to, interrogate, and extend the mapping of SHCI over time. Through this poster paper, we aim to strengthen our SMR and to explore

how SHCI can collectively reflect on how the field is delimited, structured, and interpreted before the map is finalized.

References

- [1] Raghad Albar, Andrea Gauthier, and Asimina Vasalou. 2024. A Playful Path to Sustainability: Synthesizing design strategies for children’s environmental sustainability learning through gameful interventions. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference*. 201–217.
- [2] Ferran Altarriba Bertran, Samvid Jhaveri, Rosa Lutz, Katherine Isbister, and Danielle Wilde. 2018. Visualising the landscape of human-food interaction research. In *Proceedings of the 2018 ACM Conference Companion Publication on Designing Interactive Systems*. 243–248.
- [3] Eli Blevis. 2007. Sustainable interaction design: invention & disposal, renewal & reuse. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 503–512.
- [4] Christina Bremer, Bran Knowles, and Adrian Friday. 2022. Have we taken on too much?: A critical review of the sustainable HCI landscape. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–11.
- [5] Hronn Brynjarsdóttir, Maria Håkansson, James Pierce, Eric Baumer, Carl DiSalvo, and Phoebe Sengers. 2012. Sustainably unpersuaded: how persuasion narrows our vision of sustainability. In *Proceedings of the sigchi conference on human factors in computing systems*. 947–956.
- [6] Laura D Cosio, Oğuz’Oz’ Buruk, Daniel Fernández Galeote, Isak De Villiers Bosman, and Juho Hamari. 2023. Virtual and augmented reality for environmental sustainability: A systematic review. In *Proceedings of the 2023 chi conference on human factors in computing systems*. 1–23.
- [7] Carl DiSalvo, Phoebe Sengers, and Hildur Brynjarsdóttir. 2010. Mapping the landscape of sustainable HCI. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 1975–1984. doi:10.1145/1753326.1753625
- [8] Lon Åke Erni Johannes Hansson, Teresa Cerratto Pargman, and Daniel Sapiens Pargman. 2021. A decade of sustainable HCI: connecting SHCI to the sustainable development goals. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [9] Naomi Klein. 2014. *This changes everything: capitalism vs. the climate* (first simon & schuster export edition ed.). Simon & Schuster.
- [10] Bran Knowles, Oliver Bates, and Maria Håkansson. 2018. This changes sustainable HCI. In *Proceedings of the 2018 CHI Conference on human factors in computing systems*. 1–12.
- [11] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372 (2021), n71. doi:10.1136/bmj.n71
- [12] Kai Petersen, Robert Feldt, Shahid Mujtaba, and Michael Mattsson. 2008. Systematic mapping studies in software engineering. In *Proceedings of the 12th International Conference on Evaluation and Assessment in Software Engineering*. 68–77.
- [13] Kai Petersen, Sairam Vakkalanka, and Ludwik Kuzniarz. 2015. Guidelines for conducting systematic mapping studies in software engineering. *Information and Software Technology* 64 (2015), 1–18. doi:10.1016/j.infsof.2015.03.007
- [14] Elizabeth Shove, Matt Watson, and Mika Pantzar. 2012. The dynamics of social practice: Everyday life and how it changes. (2012).
- [15] M. Six Silberman, Lisa Nathan, Bran Knowles, Roy Bendor, Adrian Clear, Maria Håkansson, Tawanna Dillahunt, and Jennifer Mankoff. 2014. Next steps for sustainable HCI. 21, 5 (2014), 66–69. doi:10.1145/2651820
- [16] Jacob O Wobbrock and Julie A Kientz. 2016. Research contributions in human-computer interaction. *interactions* 23, 3 (2016), 38–44.